Sahaj Bhadja

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Summary

Platform Security Engineer with 2+ years of experience building **Python services**, **authentication and post-quantum security designs**, automated tests, distributed systems, and production debugging workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**

- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant FastAPI services** with explicit state transitions, failure isolation, recoverable processing, and observable workflows under sustained production load reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and row-level security**, designing normalized schemas and tenant isolation for a production education platform supporting 500+ users reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*

- Achieved 96%+ classification accuracy across 650K+ URLs by comparing **Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models** within a reproducible evaluation pipeline for phishing and malware detection.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **security classification features** that exposed phishing and malware patterns reliably at scale.

Pandit Deendayal Energy University | *Research Fellowship* | *India* **January 2023 – May 2024**

- Designed a **distributed authentication architecture** spanning user, IoT-device, and role-based access scenarios, evaluating smart-contract verification, immutable identity records, and self-sovereign credentials to reduce reliance on centralized identity stores.
- Published peer-reviewed security research after analyzing **identity attack protections** against replay, phishing, man-in-the-middle, and credential theft, documenting access-control trade-offs and presenting technically defensible decentralized authentication designs at IEEE conferences.

Skills

Languages: Python, Java, C++, SQL, Bash, Shell, TypeScript
Security: Post-Quantum Crypto, Lattice Crypto, FALCON, Authentication, RBAC, Identity Security, Phishing, Malware
Systems & Networking: Linux, Unix, TCP/IP, Distributed Systems, Microservices, REST APIs, FastAPI, Multithreading
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, Git, GitHub, Jenkins
Quality: PyTest, Mockito, Unit Tests, Integration Tests, Automation, CI/CD, Debugging, Logging, Observability, Model Evaluation

Projects

Post-Quantum Key Management Service - Design Concept

- Designed a **post-quantum key service** concept for crypto agility, certificate rotation, migration, policy enforcement, audits, recovery, compatibility tests, and staged rollout across distributed storage systems with safeguards.

AI-Generated Code Security Harness - Design Concept

- Designed an **agent-code security harness** concept to isolate changes, scan dependencies and secrets, run adversarial tests, flag insecure patterns, record provenance, and require human approval before merge.

Quantum-Resistant Blockchain Research | Live

- Co-authored an **IEEE post-quantum security framework** combining FALCON lattice signatures, DPoSB consensus, behavior monitoring, and Borda Count witness selection after analyzing Shor and Grover threats to blockchain authentication.